

Towards an Augmented Reality Agent to Support Communication for Nonspeaking Autistic People

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ABSTRACT

Many nonspeaking autistic people have learned to communicate using a letterboard. With this technique, a trained Communication and Regulation Partner (CRP) holds a plastic letterboard and reads aloud the letters an individual is pointing toward. Training to communicate in this manner is typically a lengthy process and the cost and availability of a professional CRP can be a limiting factor to an individual's ability to progress. To address this problem, we build a system that allows an individual to wear an Augmented Reality (AR) headset and engage in spelling exercises on a holographic letterboard supported by a fully embodied, virtual CRP. Specifically, we video recorded 20-minute spelling sessions collected from 13 participants and coded these videos to understand the nature of interactions between a user and their CRP. Later, we use the coded data to derive a Markov decision process behaviour model that allows the virtual CRP (ViC) to mimic a real CRP. Our preliminary results show that ViC can achieve an accuracy of up to 68% even with the limited data we were able to collect and the simplified modeling approach we used. Thus, our work opens doors for nonspeakers to practice various aspects of spelling-based communication in the absence of a human CRP, thereby fostering increased independence and privacy.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction paradigms**; **Collaborative interaction**;

KEYWORDS

Nonspeaking Autistic People, Accessibility, Augment Reality, Virtual Assistant

ACM Reference Format:

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1 INTRODUCTION

About 1.5% of the world's population is diagnosed with Autism Spectrum Disorder (ASD) [17]. Approximately 30% of autistic individuals are nonspeaking [24]. In addition to acting in socially non-normative ways and having restrictive and repetitive behaviors that are the hallmarks of autism [2, 15], nonspeaking autistic people cannot communicate effectively using speech. Unfortunately, most nonspeaking autistic people are never provided access to an effective language-based communication alternative. Not having a way to convey information or share their thoughts severely restricts their educational, social, and employment opportunities.

Some nonspeaking autistic people have received specialized training that has allowed them to learn to communicate by pointing to letters on a letterboard with the assistance of a trained Communication and Regulation Partner (CRP) [9, 21] (see Figure 1). Nonspeaking autistic people who communicate using a letterboard have graduated from college (e.g., [3, 5]) and written best-selling memoirs [8]—achievements that would not have been possible if they had not learned to communicate in this way.

Learning to communicate using a letterboard is a lengthy process. In a typical training session, the CRP delivers a customized lesson that involves age-appropriate educational content. Periodically, the CRP asks the student questions about the material, inviting them to spell their responses. Most students begin spelling on three stencil boards, each with a subset of the alphabet. As they become more proficient, they learn the motor skills required to be able to isolate and point to letters on a laminated letterboard with the entire alphabet as shown in Figure 1.

The CRP has at least three roles in these interactions. First, they monitor the speller's attention to ensure they are focused on the task at hand, often verbally prompting the user to engage with the letterboard (e.g., to spell the word "HELLO", the CRP might instruct the nonspeaker to find each letter on the board - if it's on the left, right, top or bottom, etc.). Second, they provide positive feedback, encouraging spellers to continue to spell [21]. Third, since many nonspeaking autistic people move about frequently [8], the CRP may reposition the letterboard in their field of view.

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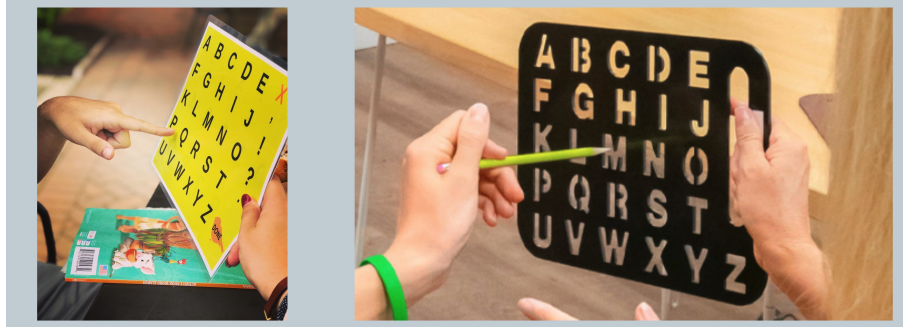


Figure 1: Training using physical letter boards. The left panel shows a laminated paper board while the right one shows a plastic stencil board.

Image sources: Growing Kids Therapy Center, <https://growingkidstherapy.com/shop/>, accessed February 20, 2024. Image source: Inside Voices, <https://www.aalive.org/spelling-to-communicate>, accessed February 20, 2024.

Many nonspeakers have used this method as an intermediary to learn to independently type on other devices such as keyboards or tablets [11]. These types of electronic devices can afford users many more opportunities, e.g., online communication and enhanced privacy. However, achieving more independence requires regular opportunities for practice. As with other skills that require consistent practice supervised by a trained professional, the cost and availability of a professional CRP can be a limiting factor to a student's ability to progress. Furthermore, in our consultations with the community, most nonspeakers express a strong desire for greater autonomy and described how reliance on their CRP can complicate social interactions where privacy would be preferred, i.e., conversing with friends or obtaining medical treatment. Recently, during the COVID pandemic, many services transitioned online, making in-person appointments between a CRP and nonspeaking individuals difficult. The assistance of a virtual CRP could be a useful asset to the nonspeaking community. Thus, we believe that a well-designed virtual agent could assist nonspeaking autistic people in learning to use a letterboard and could provide assistance when a trained professional is unavailable.

In this work, we describe the design and implementation of a realistic, embodied virtual agent developed to mimic the behaviour of a professional CRP using an empirically derived behaviour algorithm. We call this virtual CRP agent ViC - short for **Virtual CRP**. ViC interacts with a student in real-time, via an augmented reality (AR) headset, providing the student with feedback including corrections, encouragement, and positive reinforcement based on the student's interactions with a virtual letterboard. ViC was designed to specifically mimic the behavior of human CRPs based on observations from video data recorded during 13 different student sessions. These sessions were systematically reviewed and thematic analysis was used to determine salient interaction categories between students and their CRPs. Then, the videos were reviewed again to analyze the behavioural antecedents exhibited by the students as well as the consequences of these interactions with the CRP. This method was chosen to mimic the Antecedent-Behaviour-Consequent (ABC) model commonly used to assess behaviour patterns in therapeutic

settings. We use the antecedent-behaviour transitions to establish a Markov Decision Process (MDP) that embodies the ABC interactions derived from the videos. In this manner, the student's antecedent interaction can be thought of as the state of the MDP, and the CRP's selected behavioural intervention is the action. The consequence then represents the next state in the sequence (i.e., the next antecedent for the next behavioural intervention). These ABC interactions, along with the aggregate statistical data, are used to drive the behaviour algorithm for the ViC agent.

To validate our design, we conducted a Monte Carlo simulation. Our results indicate that the behaviour algorithm underlying ViC is able to achieve an accuracy of up to 68% even with the limited data we were able to derive via qualitative analysis and the simplified "memoryless" modeling approach underlying the MDP we employed. This suggests that ViC would allow a nonspeaking student to practice spelling exercises at their own convenience, complementing the training received with a professional CRP. We believe our current implementation can therefore be used as the basis for developing a more complex agent that leverages artificial intelligence and reinforcement learning to mimic other and more complex types of support provided by human CRPs (e.g., automated practice opportunities for developing open-ended communication). In this manner, we hope to eventually empower nonspeakers with more independence and privacy in their communication.

2 RELATED WORK

Augmentative and Alternative Communication (AAC) systems are intended to provide people with speech-related disabilities access to other forms of expressive communication. For nonspeaking autistic people, some commonly used AAC systems are tablet-based apps that allow users to tap icons to make requests or comments [16, 18]. These systems can be useful for making requests in particular. But they are limited in that the vocabulary available to a user is chosen by someone else. Furthermore, to be able to use AAC apps, users must first learn the specific skill to isolate and point to a symbol [25], which can be difficult for nonspeaking autistic people, many of whom have difficulty with fine motor skills [7, 13, 14, 22].

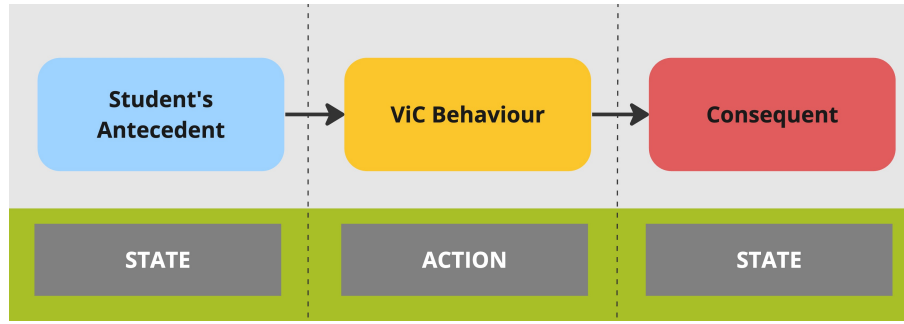


Figure 2: Illustration of the Antecedent, Behaviour, and Consequent triad with how it relates to the Markov decision chain.

Immersive technologies, such as AR, have considerable potential for providing users with opportunities to practice the motor skills involved in interacting with highly customizable communication interfaces (including keyboards). For instance, AR interfaces can be dynamically rescaled as necessary, becoming much larger than a traditional screen could accommodate and adopting as the user’s skills improve. Additionally, AR interfaces can be designed to automatically follow an individual’s movements, which could be especially advantageous for this population given the importance of movement for self-regulation in autism [8, 10]. Thus, not having to rely on someone else to hold a letterboard or tablet, e.g. a CRP or family member, has the potential to give nonspeakers more autonomy.

Despite their potential, immersive technologies in the autism space have typically not been designed for (or tested with) nonspeaking autistic people. A number of assumptions may have prevented researchers from considering the utility of head-mounted AR for this population, including concerns that sensory sensitivities might prevent them from being able to tolerate the device and concerns that they would not understand or be able to learn how to interact with holographic objects. However, recent work has shown both of these concerns to be unfounded. Nazari et al. [20] showed that 15 of the 17 nonspeaking adults tested could tolerate wearing the Microsoft HoloLens 2 head mounted device (HMD), and all 15 of them learned to interact with holographic buttons, engaging in button-pushing tasks of increasing difficulty. Building on this work, Alabood et al. [1] conducted a study showing that 5 nonspeaking autistic individuals who used a physical letterboard to communicate with assistance from a CRP could spell short words on a virtual letterboard in an AR environment with much less CRP support (see also [12, 23]).

Virtual agents have been used to assist individuals with ASD improve their communication and social skills previously (e.g. [4, 6]). However, this previous work has been mostly focused on improving nonverbal communication cues and pro-social behaviours. In the current work, we extend this approach to AAC systems for nonspeaking individuals, leveraging their pre-existing skill sets in a new, immersive environment, by developing an automated ViC agent. This ViC agent can provide a user in the AR environment with the same kinds of motor coaching, attention redirection, and encouragement as a human CRP, thereby paving the way for increased independence.

3 METHODOLOGY

3.1 Application design

Following a user-centred design approach, we developed a customized holographic spelling application in consultation with community stakeholders. The application was built for the Microsoft HoloLens 2 AR headset using the Unity game engine and the Microsoft Mixed Reality Toolkit (MRTK). In order to simulate the exact conventional experience between a user and their CRP, we 3D modeled three of the most common letterboards used by nonspeakers. These models were later imported into Unity to add the required interactions (hovering over, pressing, releasing) as well as sound and visual effects (highlighting selections and button animations). Furthermore, we developed the application to be a shared experience where both users (the student and their CRP) can interact with the holographic letterboard simultaneously. Microsoft Azure Spatial Anchors were used to synchronize virtual letterboards amongst all connected devices in the same real-world location.

3.2 Data collection

With the help of nonspeaking self-advocates, psychologists, speech language pathologists, occupational therapists, and CRPs, we developed a 20-minute training module for our user study (Figure 3). The training module consists of 5 phases of holographic spelling training. Phase 1 involves interacting with large holographic buttons to ensure the user can see the interface correctly and can successfully interact with the virtual objects. Phase 2 consists of spelling single letters on a physical letterboard, followed by spelling the same letters on the holographic one, a common technique called “mirroring”. This technique is used to acclimatize users with the virtual letterboard by providing an explicit connection to their experience using a physical letterboard. Phase 3 extends Phase 2 by increasing the difficulty with spelling complete words on the virtual letterboard, but providing additional highlighting visual cues for the correct letter selections. Phase 4 continues with spelling full words on the virtual letterboard, but without the additional visual cues. Finally, Phase 5 tests the user’s ability to interact with the virtual letterboard without first interacting with the physical letterboard.

CRPs were instructed to follow the training module, which is integrated into the holographic spelling application. Typically, CRPs

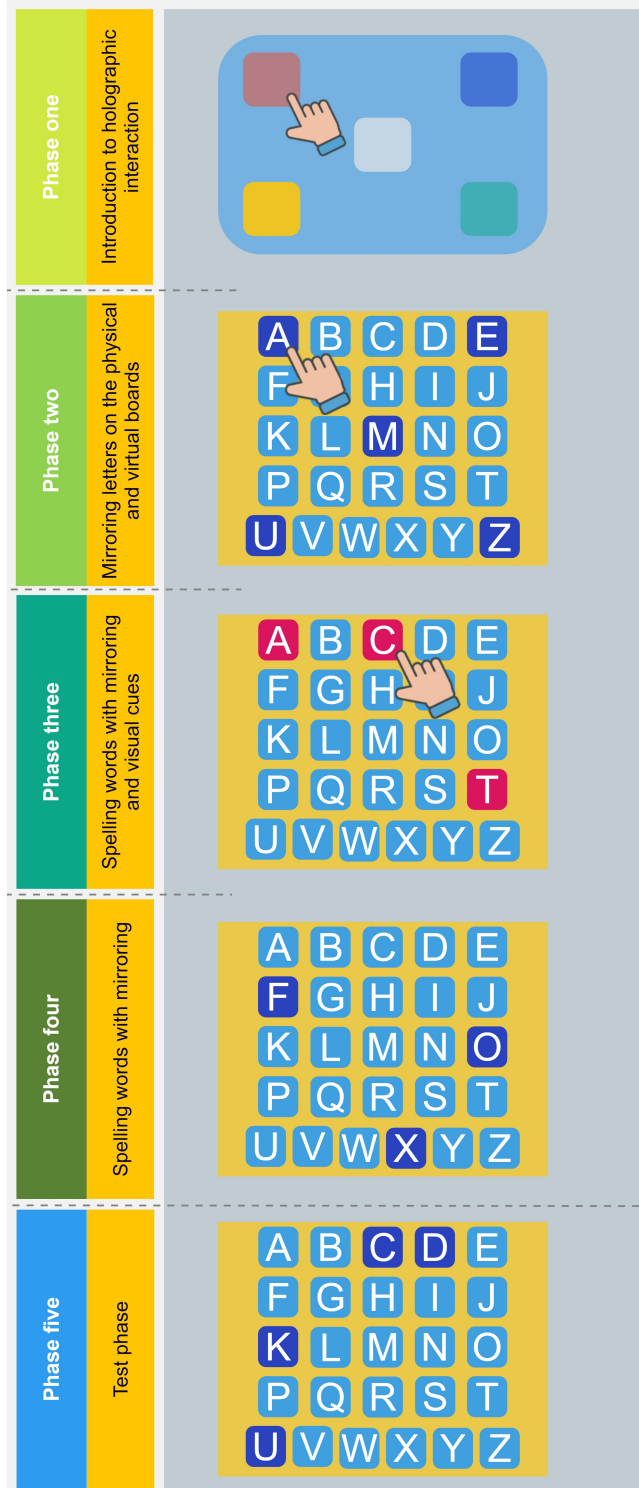


Figure 3: Diagram illustrating the data collection procedure. Phase 1 was interacting with coloured virtual buttons as an introduction. Phases 2 through 4 involved using a physical letterboard first for mirroring assistance. Phase 5 used the virtual letterboard only.

would provide different types of interventions during a conventional spelling session. These interventions range from gently waving the letterboard to restore the student’s attention, to providing detailed verbal prompts. We asked the participating CRPs to continue practicing these interventions when required throughout the study but after first giving the user a chance to complete the task unassisted. We recorded video data from the user’s HoloLens 2 device, the CRP’s device, and, in some cases, from an external mobile device connected as a client to the system. We use the collected qualitative video data later for performing thematic analysis to identify and categorize the CRP interventions, the antecedent student action, and the consequent outcome of the intervention. We then use this data to calculate the aggregate quantitative statistical values for specific CRP behaviours during the sessions which we then use to generate a MDP behaviour algorithm for our ViC agent.

3.3 Study participants

For our study, we collaborated with nonspeaking community members, including CRPs, to recruit 13 participants meeting specific criteria. Specifically, participants had a formal autism diagnosis, were at least 13 years old, unable to communicate solely through speech, and had a minimum of 2 years of physical letterboard training. The study took place in three U.S. cities, and sessions were conducted by two of the researchers. Participant ages ranged between 13 and 30 years old, with a range of letterboard experience between 2.25 and 4 years. Three participants (23%) were female, consistent with the 4:1 male-female ratio in autism diagnoses reported in the literature [26].

3.4 Data Analysis

The goal of the data analysis is to identify, categorize, and count user interactions in order to develop an empirically derived CRP behaviour model which can be used when designing our ViC agent. For this purpose, we coded video data collected from 13 participants in our user study. Qualitative analysis was carried out to process the interactions while statistical analysis was performed to limit any subjective bias. Therefore, the qualitative video data was reviewed by authors of this paper individually using NVivo 14 software. Thematic analysis was used for analyzing the video data. The video coding procedure was as follows: First, all instances of CRPs interacting with their nonspeaking students were coded. An interaction was defined as any verbalization from the CRP to the student. Even though repositioning of the letterboard is one of the CRP’s responsibilities during sessions, we decided to limit our model to verbal interactions only, as position in 3D space is hard to deduce from video data alone. Future work could involve accurately measuring position data using the HMD sensors and/or other wearable devices. After coding all videos, the videos were reviewed again to code the five-second intervals both preceding and following the documented interaction as the Antecedent and the Consequent events, respectively. We chose to use five-second intervals as we wanted to allow enough time for an interaction to occur completely, but also to limit the time such that a Behaviour could reasonably be casually linked to the preceding Antecedent. The relationship and flow from the Antecedent to Consequent was used to cluster the codes into themes representing implicit types.

Table 1: Probability Matrix. Each value represents the probability of the specific behavioural intervention being chosen given that the specified antecedent event occurred.

Antecedent	Encourage	Prompt	Positive Reinforcement
Complete task	0.0	0.0	1.0
Correct selection	0.22	0.16	0.62
Hover over correct selection	0.67	0.33	0.0
Hover over incorrect selection	1.0	0.0	0.0
Incorrect selection	0.56	0.38	0.06
No interaction	0.34	0.64	0.02
Start new task	0.25	0.50	0.25

This resulted in the set of all Antecedent, Behaviour, Consequent triads (ABC triads).

4 RESULTS

In total, 454 interactions were coded. Each interaction was clustered into one of three emergent themes: Encouragement, Prompt, and Positive Reinforcement. Encouragements were defined as any interaction where the CRP provided assistance to the student *without* explicitly specifying the next correct letter selection or aiding them to follow through with the task at hand. A Prompt was defined as any interaction where the CRP provided assistance that included explicitly stating the next correct letter selection or specific instruction for the task. Positive Reinforcement was defined as any interaction where the CRP did not provide assistance with a task, but instead expressed a positive morale boost during or after the fulfillment of the current task. To evaluate the reliability of the coding process, inter-rater reliability was measured using Cohen's Kappa - which measures agreement due to chance - for randomly selected sections of videos. The values of Cohen's Kappa for the themes were: Prompt 0.69, Encouragement 0.68, Positive Reinforcement 0.74, thus signifying substantial agreement between raters. Fifty of the interactions were categorized as Instructions, i.e. initially directing the student to perform a specific task. These interactions were discarded from further analysis as instructions are necessarily task specific. Out of the 404 interactions considered, 20% were set aside for validation testing using stratified sampling to preserve the classification proportions and the rest were coded and selected for the next step of the process - determining their Antecedent and Consequent to form an ABC triad.

Each ABC triad can be thought of as a logical If-Then statement and an outcome - i.e., if the student performs Antecedent A, then the CRP performs Behaviour B which results in Consequent C (the outcome of the intervention). To formulate this algorithm as a MDP - a requirement for some forms of reinforcement learning - we need to be able to select a Behaviour using only the current state (the Antecedent). To establish a Markov chain, we can define the Consequent to be the next state in the chain. This creates a decision making system where the Antecedent (input) is used to predict a Behaviour (output) and the Consequent of that Behaviour becomes the next Antecedent in the chain (illustrated in Figure 2). Because each Antecedent and Behaviour belong to a limited set of possible categories, there is a finite number of combinations. Therefore, it is possible to calculate the conditional probability of

selecting Behaviour B given Antecedent A has occurred (see Table 1). Using these conditional probability values we can calculate a probability matrix which can be used to quantitatively determine the Behaviour for the ViC model to execute.

5 VIRTUAL CRP IMPLEMENTATION

We embodied a virtual CRP agent (the ViC) into our initial holographic spelling system. The ViC was designed and animated using Character Creator 4, iClone 8, and Unity's animation state machine architecture. When we designed the ViC agent we wanted to emphasize the photo-realism of the agent as much as possible, as working with a human CRP is an essential element to the letterboarding technique [21]. We believe that in order to replicate this effectively, we need to provide a similarly realistic experience for the speller in the virtual environment. Furthermore, we also wanted to include verbal and visual feedback in a similar way that a real, human CRP would. To this end, we leveraged the AccuLips functionality of iClone 8 to automatically generate viseme-based speaking animations for the agent. Further, we gave the agent a voice by utilizing the Microsoft Azure Speech SDK, a cloud-based service that allows for real-time speech-to-text synthesis. To facilitate facial animation synchronization, all speech was pre-generated using the service API and recorded as MP3 files to be played on demand during application runtime. In the future, we hope to expand the realism of the ViC agent by integrating real-time text generation and text-to-speech audio creation.

To bring the ViC agent to life, we created a custom behaviour tree data structure using Unity's C# scripting backend. Behaviour trees are commonly used in modern video games to handle complex Non-Player Character (NPC) behaviours. Every frame of the application, the behaviour tree is traversed and the appropriate logic is executed. For our current prototype, the behaviour tree handles the ViC agent's animation and speech logic. To add variation to the behaviours of the ViC agent we included fuzzy logic calculations for selecting individual animations and audio files for playback. For example, the positive feedback behaviour audio output is randomized such that a variety of potential positive feedback audio files can be selected and played to create a more diverse and realistic experience.

To manage the application itself, we designed a custom Finite State Machine (FSM) architecture. A FSM architecture is a computational model used to design systems that can be in only one of a finite number of states at any given time (e.g. a traffic light



Figure 4: Screenshots of the ViC agent and the letterboard interface on the Microsoft HoloLens 2 headset.

system). The FSM consists of 4 states and 5 possible transitions between states. Initially, the FSM initializes the application into the introductory state, where the ViC agent greets the user, explains their role as a virtual CRP, and informs the user of the purpose of the application. To transition from this state to the next, the user must provide a thumbs-up gesture in the direction of the ViC agent as per the agent's instruction and demonstration. To accommodate this function, we utilized the MRTK hand joint recognition library. Using this, we are able to define custom gesture positions based on the position and orientation of the user's hand and fingers in the space. Following this, the FSM transitions into the word generation state, where a random word is selected from a predefined list of words that can be imported as a CSV file. Once a random word has been selected, the FSM transitions to the spelling state. In the spelling state the agent awaits input from the user via the custom virtual letterboard interface and acts according to its behaviour algorithm. This is when the pre-generated probability matrix is referenced to quantitatively determine what the ViC agent will select as its behaviour given the student's input. Most of the time spent in the application occurs in this state. This state can transition back to the word generation state in one of two distinct ways: successfully, when the student correctly spells the randomly selected word, or unsuccessfully, when the ViC agent determines too many incorrect attempts have been made. Further iterations of this application will likely need to include an additional end of state trigger to allow for a new word to be selected, for instance, if the student wishes to skip the current word. Finally, once all of the words imported from the CSV file have been attempted, the FSM will transition to the end state where the ViC agent congratulates the user for finishing and bids them farewell before the application is terminated.

Due to computational limitations of the HoloLens 2, rendering the ViC agent realistically was challenging without sacrificing the device's performance. To address this, the application was designed to utilize Holographic Remoting on a dedicated machine with compatible hardware, achieving real-time frame rates and incorporating advanced features such as hair and cloth physics simulation. Although direct audio output from the HoloLens 2 was not viable with

Holographic Remoting, audio redirection to Bluetooth headphones on a laptop can be leveraged for human trials.

Finally, we created data capturing mechanisms during the application runtime to collect user data such as hand positions, hand velocity, gaze direction, user position, and user velocity. This data is collected in addition to metrics such as the number of correctly and incorrectly spelled letters, time spent spelling, and the number and the type of ViC interactions. Our hope is to use this data to iterate on our design and improve the behaviours of our agent based on user feedback.

6 VALIDATION

To validate our behaviour model, we tested - using a Monte Carlo simulation - the generated probability matrix against 20% of the data that was left out of the previous data analysis steps via stratified sampling. A Monte Carlo simulation is a computational technique used to estimate the probability of different outcomes by running multiple random trials to model their probability in a process or system. The test data was coded via the same procedure as the previous data. First, the behaviours were coded and categorized into three themes. Second, the Antecedent and Consequent events were coded. Then, the Antecedent data was fed into the probability matrix and used to calculate the predicted behaviour of the ViC model. This process was simulated 1000 times to generate a random sample distribution of outcomes. The accuracy of the behaviour model was then calculated by determining the successful predictions (i.e., the same as the coded behaviour of the human CRPs in the test data) out of the total number of predictions in each simulation. The sample mean ($\bar{x}$) was 55.25% with a standard deviation of 3.61%. The minimum prediction accuracy in these simulations was 45.00% and the maximum was 68.64%. To determine if our model behaved better than chance (null hypothesis $H_0: \bar{x} = 33\%$) we performed a t -test and computed a p -value < 0.01 (95% CI: 55.25% $\pm$ 0.2240%). Given the limited number of interactions used to build the model and the current simplicity of the model, these results are promising and point to the feasibility of incorporating realistic behaviours into the ViC agent.

7 DISCUSSION

It is important to note that the described behaviour model is a simplified representation of what is in reality a complicated interpersonal communication and social behaviour. Thus, the intention of this model is not to achieve maximal accuracy at predicting CRP behaviour. We are interested in determining whether we are able to position the behaviour of our ViC agent as a memoryless MDP. This means that we can achieve an accuracy greater than chance even though we are limited to decision making using only the current state of the environment - in this case the Antecedent student interaction. This is important, because it will allow us to use this behaviour model as a baseline for future models that leverage artificial intelligence and reinforcement learning to optimize the state-action policy (i.e., selecting the best possible behaviour as an action given the current state of the student's interaction). We intend to leverage the model generated in this work as the entry point for another user trial study where students work with an online reinforcement learning agent in real time, allowing it to improve iteratively as they continue to work together over time. Similarly, this will allow the behaviour algorithm to adapt to individual user experiences. For example, some users may benefit from more positive reinforcement while others may prefer less. Given every autistic individual's experience is unique, we believe an adaptive and customizable algorithm will be integral to promoting a successful and positive user experience.

As mentioned previously, it is commonly believed that working with a human CRP is an essential element to the letterboard communication technique [21]. We hope that, by providing a sufficiently realistic virtual facsimile and behaviour model, we can glean insight into the relationship developed by a student and CRP. Our eventual goal is to augment and enhance the training offered by human CRPs via the use of virtual agents. One of the benefits of working with an autonomous ViC agent in a virtual environment is we are able to collect a variety of data that is continuously monitored via the sensors on the head mounted devices. Hopefully this data can be used to further illuminate best practices when working with individuals in this population.

Several limitations are present with this early prototype. First, one of the CRP's responsibilities is the dynamic repositioning of the letterboard when they work with a student to help accommodate for the frequent movement many nonspeaking autistic people exhibit [8]. Unfortunately, we were not able to address this aspect of CRP behaviour in this work. As movement of the virtual letterboard in 3D space requires real-time control of 6 continuous variables (i.e., position and rotation). We believe a second model, which controls 3D positioning, that works in concert with a behavioural model similar to the one described in this work may be necessary [19]. Second, while we believe that a realistic human model will perform best in lieu of a real, human CRP, further research validating these assumptions should be conducted. Moreover, it may be the case that user experience could be improved by increasing the realism of certain aspects of the virtual agent while decreasing others. Finally, during our data analysis, we elected to use windows of 5 seconds for coding the Antecedents and Consequents. We chose this duration as it provided a reasonable trade-off between allowing for enough time for an event to occur, but not too long as to question the causality

of the events. Therefore, an investigation that examines multiple different duration windows and compares the results would be beneficial.

8 CONCLUSIONS

In this paper we have laid out a framework for creating a photo-realistic, virtual assistant designed to help nonspeaking autistic individuals communicate via a holographic letterboard. To the best of our knowledge, this is the first system that allows a nonspeaker to interact with an embodied virtual agent in lieu of a real human. Our preliminary results suggest that this a feasible approach. We hope to further extend this system using the data collected from the described application and a subsequent user study to create an improved version that models user behaviour in real-time by leveraging more advanced behaviour modeling techniques. In the longer term, we believe that our system would afford more independence and privacy to the nonspeaking population by reducing their reliance on human CRPs.

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